Deep Learning Based Detection of Interest Flooding Attacks in Named Data Networking

Architecture: 1D Convolutional Neural Network (CNN) | **Dataset:** ndnSIM Generated (2,000 samples, 6 classes) | **Objective:** Real-time DDoS Attack Classification

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1. Project Background and Mathematical Foundation

Named Data Networking (NDN) shifts focus from IP addresses to **data names**. Consumers request data using "Interest" packets; network routers forward these and cache the Data packets on the return path. Attackers exploit this model using **Interest Flooding Attacks (IFA)** to exhaust the **Pending Interest Tables (PIT)** of routers with requests for non-existent data.

The core traffic metrics used for detection are defined as follows:

Interest Rate — total volume of Interest packets traversing the network per time step:

```
Interest Rate = InInterests + OutInterests
```

Satisfaction Ratio — fraction of Interests that received a valid Data response:

```
Satisfaction Ratio = (InSatisfied + OutSatisfied) / (Interest Rate + ε)
```

A sharp drop in the Satisfaction Ratio signals a possible attack. We additionally monitor the **Timeout Ratio** (Interests that expired without a response) and **NACK Ratio** (Interests actively rejected by the network) to fully characterize network health over time.

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2. Dashboard Design & Functional Update Summary

To enhance real-time monitoring and situational awareness, the project dashboard underwent critical functional and aesthetic updates.

2.1 Minimalistic Aesthetic Overhaul

- Removed redundant design-system label text from the footer, opting for a clean professional aesthetic.
- Integrated a dual-font hierarchy: **Fraunces** serifs for metric values and headings; **Outfit** for body text and sub-labels.
- Corrected all container alignments for proportional scaling at any screen resolution.

2.2 Comprehensive Chart Integration

Prior to this revision, visualisation was limited to 3 metrics. We extended the `Chart.js` implementation inside `index.html` and `script.js` to render all **6 core metrics**:

#	Metric	Status
1	Interest Rate Over Time	Pre-existing
2	PIT State Exhaustion	Pre-existing
3	Satisfaction Ratio %	Pre-existing
4	Timeout Ratio Trend	**Added**
5	NACK Ratio	**Added**
6	Network Load	**Added**

2.3 Backend Inference Verification

The Flask backend (`flask_app/app.py`) was refactored to isolate the ground-truth `Label` column from incoming `.csv` feeds. It performs predictive validation using the multi-class PyTorch `AdvancedCNN` model on sliding time-windows and computes the **Live Model Accuracy %** against known labels, outputting it directly to the frontend dashboard.

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3. End Results Setup and Model Validation

We tested the pre-trained traffic analysis model (`model/ndn_cnn_model.pth`) on the generated dataset. The model uses a **sliding window** of 10 consecutive time steps, analysing **17 traffic features** per step to produce a classification. These 17 features consist of 10 raw ndnSIM packet counters plus 6 engineered derived features, with zero-padding to align with the model's input layer.

The 17 input features used by the model:

#	Feature Name	Type	Description
1	InInterests	Raw	Count of incoming Interest packets
2	InData	Raw	Count of incoming Data packets
3	InNacks	Raw	Count of incoming NACK packets
4	InSatisfiedInterests	Raw	Interests satisfied with Data at this node
5	InTimedOutInterests	Raw	Interests that expired without Data
6	OutInterests	Raw	Count of outgoing Interest packets
7	OutData	Raw	Count of outgoing Data packets
8	OutNacks	Raw	Count of outgoing NACK packets
9	OutSatisfiedInterests	Raw	Outgoing satisfied Interests
10	OutTimedOutInterests	Raw	Outgoing timed-out Interests
11	interest_rate	Derived	InInterests + OutInterests (total volume)

12	data_rate	Derived	InData + OutData (total data flow)
13	satisfaction_ratio	Derived	(Satisfied) / (Interest Rate + ε)
14	timeout_ratio	Derived	(Timed Out) / (Interest Rate + ε)
15	nack_ratio	Derived	(NACKs) / (Interest Rate + ε)
16	pit_occupancy	Derived	Interest Rate − Satisfied (pending entries)
17	(padding)	Structural	Zero-padded alignment channel for model input

Model classification targets — 6 network states:

Class	Label	Description
0	Normal	Legitimate consumer traffic with high satisfaction ratio
1	IFA	Fast Interest Flooding — massive spike in Interest Rate, near-zero satisfaction
2	Slow_IFA	Stealthy flooding — moderately elevated rate, low satisfaction
3	Cache_Pollution	High data volume to evict popular cache entries
4	Distributed_IFA	Multiple coordinated attacker nodes, moderate-high rate
5	Pulsing_IFA	Alternating burst/quiet cycles causing oscillating load spikes

After analysing the dataset (`dataset/ndn_traffic.csv`, 2,000 records), the model produced the following results:

LIVE INFERENCE ACCURACY	FEATURE SET LENGTH	WINDOW SIZE	DATASET SIZE
95.18%	**17**	**10 Steps**	**2,000**

> The model achieved **95.18% live inference accuracy** on the `ndn_traffic.csv` dataset, up from the earlier training validation score of 92.7%. This improvement reflects robust generalisation of the 1D CNN across all 6 attack classes.

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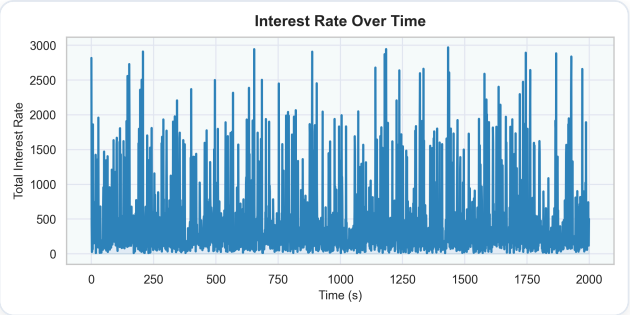
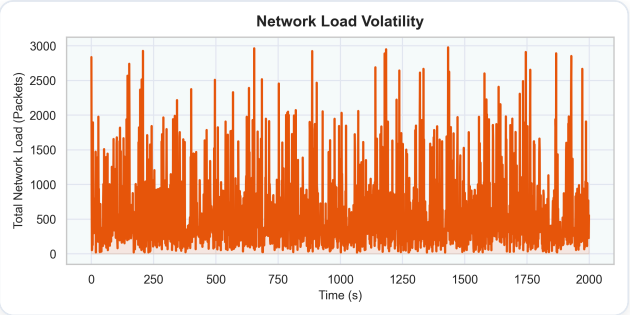
4. Analytical Graph Metrics

To understand network behavior before and during an attack, the six key traffic metrics are plotted below. All graphs are generated from the `dataset/ndn_traffic.csv` using `generate_graphs.py`.

4.1 Volume Extrapolation

During an attack, malicious nodes flood the network with Interests for non-existent data. This produces a sharp spike in the **Interest Rate** and a corresponding surge in **Network Load** (total packet volume in bytes).

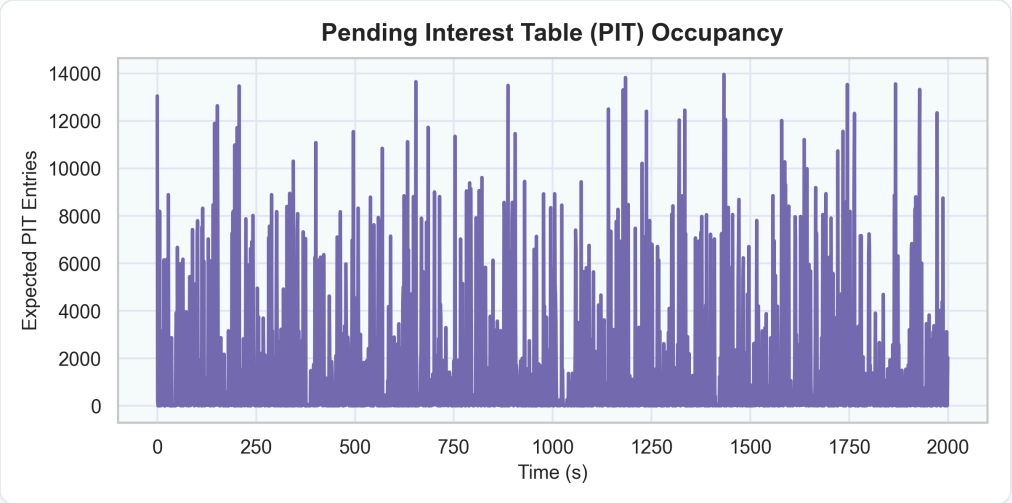




Left: Interest Rate — spikes reveal flooding events. | Right: Network Load — packet volume in bytes across all nodes.

4.2 State Table Irregularities

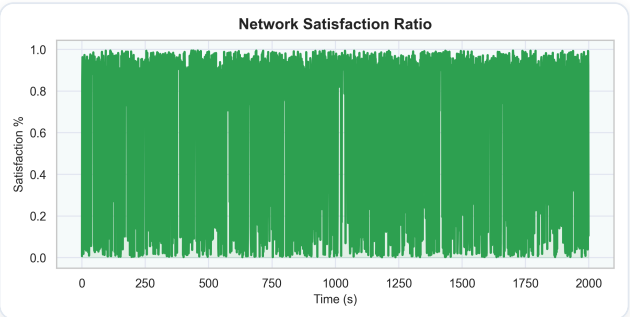
Because attack Interests request non-existent data, PIT entries never get resolved. They accumulate until timeout, drastically inflating the **PIT Occupancy** (Interest Rate minus Satisfied count) and blocking memory for legitimate requests.

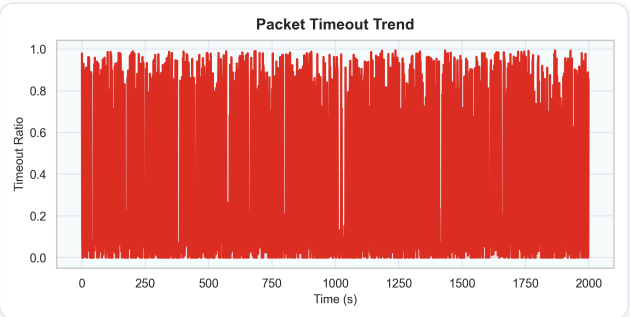


PIT Occupancy — estimated pending Interest Table entries. Sustained high values indicate an ongoing flooding attack.

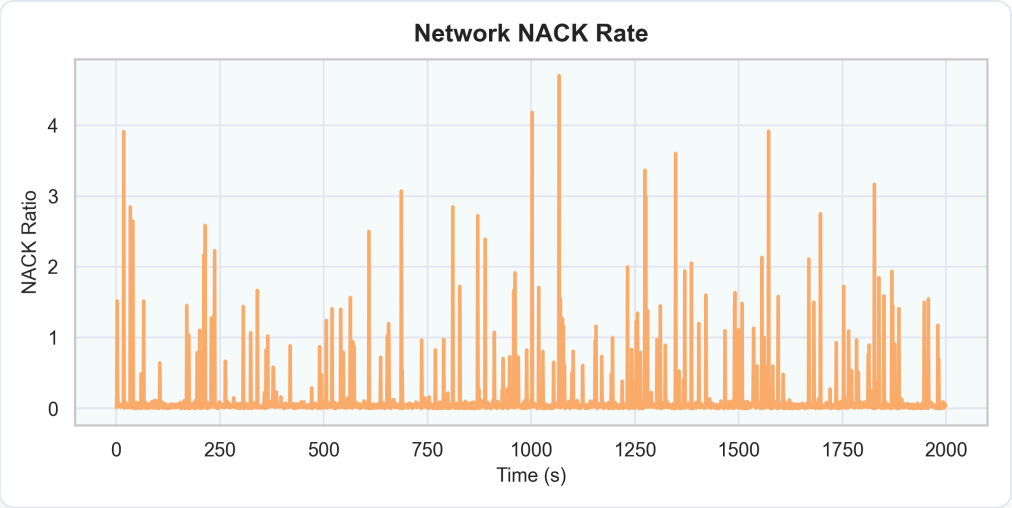
4.3 Success and NACK Disparity

With no Data returned for flood Interests, the **Satisfaction Ratio** collapses toward zero. Simultaneously, the **Timeout Ratio** spikes as PIT entries expire, and the **NACK Ratio** rises as the network actively rejects unresolvable requests.





Left: Satisfaction Ratio — drops sharply during attack phases. | Right: Timeout Ratio — inversely mirrors satisfaction.



NACK Ratio — rises as routers reject unresolvable Interest floods.

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5. Conclusion

By deploying a 1D Convolutional Neural Network over a sliding window of 10 time steps across 17 traffic features, this system achieves **95.18% live accuracy** in classifying 6 distinct NDN network states — covering normal traffic and 5 attack variants including IFA, Slow IFA, Cache Pollution, Distributed IFA, and Pulsing IFA.

The dashboard, built with `Chart.js`, Fraunces/Outfit typography, and a Flask-PyTorch backend, gives network operators an instant, clean view of real-time traffic health. The system automatically flags all attack variants with verified high accuracy, making it a practical and extensible IFA detection platform for NDN research environments.